PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME	

CT GROUP	1	6	S			INDEX NUMBER					

BIOLOGY — 9744/04

Paper 4 Practical — **29 August 2017**

Junior College Year 2 — **2 h 30 min**

READ THESE INSTRUCTIONS FIRST

Write your name, CT group and index number on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen.
You are to use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part question.

Shift	
Laboratory	
For Examiner's Use	
1	19
2	22
3	14
Total	**55**

1 You are required to investigate the glucose concentrations of solutions **S1** and **S2**.

Doctors use the analysis of urine to help diagnose some medical conditions. One such medical condition is diabetes which results in glucose being released in urine if the condition is untreated.

You are provided with:

- 15cm^3 of 4% glucose, labelled **G1**
- 10cm^3 of unknown glucose concentration representing urine, labelled **S1**
- 10cm^3 of unknown glucose concentration representing urine, labelled **S2**
- 50cm^3 of distilled water, labelled **W**
- 40cm^3 of Benedict's solution, labelled **Benedict's solution**

Proceed as follows:

1 Carry out a serial dilution of the 4% glucose, **G1**, to reduce the concentration of glucose solution by half between each concentration of four successive dilutions, to give **G2**, **G3**, **G4** and **G5**. You will also need to set up a control, **C**.

You are required to make up at least 5cm^3 of each concentration of glucose solution in the small glass vials provided.

Complete Table 1.1 to show how you will make the concentrations of the glucose solutions, **G2**, **G3**, **G4** and **G5**, and show how you will set up the control, **C**.

Table 1.1

	G1	**G2**	**G3**	**G4**	**G5**
Concentration of glucose solution / %					
Label of glucose solution to be diluted		G1			
Volume of glucose solution to be diluted / cm^3					
Volume of distilled water , **W**, to make the dilution / cm^3					
Description of the control, **C**: ………………………………………………………………………………………………… …………………………………………………………………………………………………					

[4]

Test glucose solutions and unknowns with Benedict's solution. Excess Benedict's solution is to be added to the solutions and samples. Then heat the mixture.

You should record the time taken for first appearance of any different colour or precipitate from the blue starting colour.

The result of unknown concentration of glucose will be compared with the time taken for first appearance of any different colour or precipitate obtained from glucose solutions **G1**, **G2**, **G3**, **G4** and **G5** with Benedict's solution.

2 State the volume of Benedict's solution and the volume of the solutions (**G1**, **G2**, **G3**, **G4** and **G5**) and the unknown samples you are testing (**S1** and **S2**).

Volume of Benedict's solution: ………………cm^3

Volume of each glucose solution (**G1**, **G2**, **G3**, **G4** and **G5**): ………………cm^3

Volume of sample (**S1** and **S2**) : ………………cm^3

[2]

3 Set up a water-bath and, test each test-tube separately, test all concentrations of **G** (**G1**, **G2**, **G3**, **G4** and **G5**) and **C** for the presence of glucose. Start timing when the test-tube is placed into the hot water-bath. If there is no colour change after 300 seconds, record 'more than 300' as your result.

4 Observe each test-tube very carefully for the first appearance of any different colour or precipitate from the blue starting colour and record the timing for this change.

5 (a) State one variable, other than volume, which needs to be kept constant when you do the tests.

………………………………………………………………………………………..……[1]

(b) Describe how you will keep this variable constant.

……………………………………………………………………………………………..

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....[1]

6 Use the space below to record all your results.

[4]

7 **Repeat the procedure for S1 and S2.** Estimate the concentration of glucose in sample **S1** and **S2** by comparing with the time taken for the first appearance of any different colour or precipitate obtained from testing solutions **G1**, **G2**, **G3**, **G4** and **G5** with Benedict's solution**.**

Time taken for first appearance of any different colour or precipitate from the blue

starting colour for **S1**: ……………………………

Concentration of glucose in **S1**…………………..[1]

Time taken for first appearance of any different colour or precipitate from the blue

starting colour for **S2**: ……………………………

Concentration of glucose in **S2**…………………..[1]

[1]

8 Identify two significant sources of error in this procedure and for each, suggest how you would improve the procedure to minimize the source of error.

Source of error………………………………………………………………………...

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Improvement…………………………………………………………………………

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Source of error………………………………………………………………………

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Improvement…………………………………………………………………………

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[Total: 19]

2 During this question you will require access to

- a microscope fitted with eye piece graticule
- and slide **K2.**

K2 is a cross section of a portion of a leaf from plant that grows in full sunlight and are adapted to relatively high light intensities (sun leaf).

(a) Examine the slide under a microscope and locate a suitable cross section for your plan diagram as seen in Fig. 2.1. In your view, you should be able to observe the distinct categories of different types of cells in a leaf cross section. Choose the lens that is most suitable for viewing the cross section of the leaf in the field of view.

Please avoid the mid rib region for your plan diagram seen in Fig. 2.1

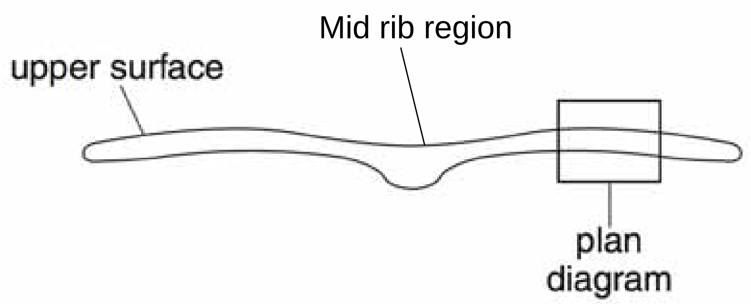

Fig. 2.1

State which objective lens you have decided to use and give a reason for your choice.

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(b)(i) Using the objective lens selected in **(a)** and the eyepiece graticule fitted into your microscope, make measurements of the total leaf thickness of **K2**.

No. of divisions of eyepiece graticule:…………………………………………..….[1]

(ii) Calculate the actual thickness of the leaf of **K2** using your data from step **(b)(i)**. Let each division on the eyepiece graticule be 0.0025mm.

Show your working with **appropriate** units.

Actual thickness of K2 …………………………………..[2]

(c) Make a detailed, labelled drawing of a section of the cross section of **K2** in the space below.

[5]

(d) (i) Table 2.2 shows the effect of light intensity on the number of chloroplasts between sun and shade plants.

Table 2.2

Light intensity / µmolm^{-2}s^{-1}	Number of chloroplasts per palisade cell	
	Sun leaf	Shade leaf
800	110	79
400	97	69
200	80	54
80	52	34
40	30	20

Plot the graph using the relevant data shown in Table 2.2. [4]

(ii) Using the graph, describe and explain a relationship between the factors investigated. [4]

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(e) In a separate study, a student recorded the stomatal density of two plants of the same species. One plant was not regularly exposed to full sunlight and often show adaptations to relatively low light intensities. The leaves of such a plant are known as shade leaves. Another plant was grown in full sunlight and are adapted to relatively high light intensities and the leaves are called sun leaves.

A statistical test was carried out to determine whether there was a significant difference in the mean stomatal density between shaded and exposed leaves.

(i) State a statistical test that could have been used to determine whether the difference in the mean stomatal density between the shade and sun leaves is significant.

……………………………………………………………………………………[1]

(ii) A summary of the student's results is shown in Table 2.3.

Table 2.3 shows the student's results.

Table 2.3

Mean stomatal density / mm^{-2}		Significance of difference	Total sample size
Shade leaves	Sun leaves		
290	335	$p < 0.05$	30

Comment on what these results show and suggest an explanation for any pattern.

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[Total: 22]

3 Fig. 3.1 shows an electrode connected to a data logger that can be used to measure the concentration of potassium ions in the water.

Fig. 3.1

Fig. 3.2

Design an experiment, using the electrode to investigate the effect of temperature from 10°C to 70°C on the permeability of potato cell membranes. Potatoes are rich in potassium ions. When small disc-shaped potatoes cut from a core borer as seen in Fig. 3.2 are placed in water, potassium ions are released from the cell into the water.

You must use:

- potatoes,
- potassium ion-selective electrode which measures in mg/L,
- a core borer of 10mm in diameter,
- distilled water.

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware .e.g boiling tubes, test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,
- blunt forcep,
- syringes,
- scalpel,
- ruler,
- timer e.g. stopwatch or stop clock,
- thermometer,
- isotonic buffer solution,
- hot water and ice.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it.
- be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total:14]

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End of Paper